



Once
you stop
learning,
you start
dying.

—Albert Einstein

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TABLE 3
DICE STATUS

Dice Face	LED1	LED2	LED3	LED4	LED5	LED6	LED7
Face1	Off	Off	Off	On	Off	Off	Off
Face2	On	Off	Off	Off	Off	Off	On
Face3	On	Off	Off	On	Off	Off	On
Face4	On	Off	On	Off	On	Off	On
Face5	On	Off	On	On	On	Off	On
Face6	On	On	On	Off	On	On	On

```
sketch_feb10a.ino
1  #define BP 4 // Button pin
2  #define L1 12 // BIT 1 (MSB)
3  #define L2 13 // BIT 2
4  #define L3 14 // BIT 3 (LSB)
5  #define ID 50 // Initial delay
6  #define FD 300 // Final delay
7  #define RS 40 // Rolling steps
8  void setup() {
9    pinMode(L1, OUTPUT);
10   pinMode(L2, OUTPUT);
11   pinMode(L3, OUTPUT);
12   pinMode(BP, INPUT);
13   Serial.begin(115200);
14 }
15
16 void loop() {
17   if (digitalRead(BP) == HIGH) {
18     int dt = ID;
19     for (int i = 0; i < RS; i++) {
20       int rn = (i % 6) + 1;
21       displayNum(rn);
22       delay(dt);
23       dt = map(i, 0, RS - 1, ID, FD);
24     }
25     int r = random(1, 7);
26     displayNum(r);
27   }
28 }
29 void displayNum(int n){
30   digitalWrite(L1, (n & 0b100) ? HIGH : LOW);
31   digitalWrite(L2, (n & 0b010) ? HIGH : LOW);
32   digitalWrite(L3, (n & 0b001) ? HIGH : LOW);
33 }
34 }
```

Fig. 6: Snippet of the source code



Fig. 7: Authors' dice prototype with LEDs

A random number generation function is set within a range of 1 to 6 to mimic a dice roll.

The correct port and board settings (e.g., NodeMCU-32S or the specific board version) must be

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Fig. 6 shows a snippet of the source code.

Construction and testing

Upload the source code into the ESP32.

Assemble the circuit on a general-purpose PCB.

Fix all seven LEDs on the PCB, including switch S1.

Refer to Fig. 7 for the prototype setup before final assembly.

Connect all ICs with the ESP32 as per the circuit diagram.

Power on the device, press and release the dice switch (S1), and observe the randomly generated number displayed using LEDs. Table 3 verifies the dice status. **EFY**

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